IDEX HT2425 + Tic T500 Controller

PC Command Reference

Document Version: 1.1 | Firmware: v2.1.7 | Updated to reflect 1/2 microstep default and corrected response strings

Connection Settings (USB to Arduino)

Parameter	Value
Baud Rate	115200
Data Bits	8
Parity	None
Stop Bits	1
Line Ending	Newline (LF) or Carriage Return (CR)

Hardware Configuration

Device	Serial Port	Arduino Pins	Baud Rate
IDEX Valve 1	Serial1	TX1=18, RX1=19	19200
IDEX Valve 2	Serial2	TX2=16, RX2=17	19200
Tic T500 Stepper	Serial3	TX3=14, RX3=15	9600

Command Format

`<device><command>[value]`

Prefix	Device
`1`	IDEX Valve 1
`2`	IDEX Valve 2
`S`	Tic T500 Stepper Motor
`?`	Help

Valve Commands

Command Summary

Command	Description	Value Required
`P`	Move to position	Yes (1–24)
`M`	Home valve	No
`S`	Query status/position	No
`E`	Read error code	No

Move to Position (P)

Moves the specified valve to a position (1–24).

Format: `<valve>P<position>`

Example	Description
`1P1`	Move valve 1 to position 1
`1P5`	Move valve 1 to position 5
`1P24`	Move valve 1 to position 24
`2P1`	Move valve 2 to position 1
`2P12`	Move valve 2 to position 12

Response:

```
> 1P5
Moving valve 1 to position 5
OK: Move accepted
```

Home Valve (M)

Returns the valve to its home position.

Format: `<valve>M`

Example	Description
`1M`	Home valve 1
`2M`	Home valve 2

Query Status (S)

Returns the current valve position or error code.

Format: `<valve>S`

Example	Description
`1S`	Query valve 1 status
`2S`	Query valve 2 status

Response:

```
> 1S
Position: 5
```

Read Error Code (E)

Returns the last error code stored by the valve.

Format: `<valve>E`

Example	Description
`1E`	Read valve 1 error code
`2E`	Read valve 2 error code

Valve Error Codes

Hex Code	Description
`0x00`	No error
`0x2C`	Data CRC error
`0x37`	Data integrity error
`0x42`	Valve positioning error
`0x4D`	Config/command mode error
`0x58`	Non-volatile memory error
`0x63`	Valve failure (cannot home)

Stepper Motor Commands

> **Default Step Mode:** The firmware defaults to **1/2 microstepping** (TIC_STEP_MODE = 1). All velocity and position values are in *microsteps*. A 200-step/rev motor requires **400 microsteps per revolution** in this mode. Velocity values must be doubled compared to full-step mode to achieve the same full-steps/sec speed.

Command Summary

Command	Description	Value Required
`SO`	Stepper ON (energize + exit safe start)	No
`SF`	Stepper OFF (de-energize)	No
`SS`	Stop (halt and hold)	No
`SV`	Set velocity	Yes (signed integer)
`SP`	Set target position	Yes (signed integer)
`SR`	Read status	No
`SC`	Set current position to 0	No
`SM`	Set max speed	Yes (unsigned integer)
`SA`	Set max acceleration	Yes (unsigned integer)

Stepper ON (SO)

Energizes the motor coils, clears any stale position/velocity target, and exits safe start mode. Must be called before the motor will move.

Format: `SO`

Response:

```
> SO
Energizing stepper...
OK: Motor energized and holding
```

Stepper OFF (SF)

De-energizes the motor coils. Motor will coast to a stop and no longer hold position.

Format: `SF`

Response:

```
> SF
De-energizing stepper...
OK: Motor de-energized
```

Stop Motor (SS)

Immediately halts the motor (ignoring deceleration limits) and holds position with coils energized.

Format: `SS`

Response:

```
> SS
Stopping stepper...
OK: Motor stopped and holding
```

Set Velocity (SV)

Sets the target velocity. Motor will accelerate/decelerate to reach this velocity.

Format: `SV<velocity>`
Units: Microsteps per 10,000 seconds
Direction: Positive = forward, Negative = reverse

> **Microstepping note:** At the default **1/2 microstep mode**, divide the SV value by 20,000 to get full steps/sec.
> Formula: *full steps/sec = SV value ÷ 10,000 ÷ 2*

Velocity Conversion Table (default 1/2 microstep mode)

Command	Calculation	Full Steps/sec
`SV200000`	200,000 ÷ 10,000 ÷ 2	10 steps/sec
`SV1000000`	1,000,000 ÷ 10,000 ÷ 2	50 steps/sec
`SV2000000`	2,000,000 ÷ 10,000 ÷ 2	100 steps/sec
`SV4000000`	4,000,000 ÷ 10,000 ÷ 2	200 steps/sec
`SV10000000`	10,000,000 ÷ 10,000 ÷ 2	500 steps/sec
`SV20000000`	20,000,000 ÷ 10,000 ÷ 2	1000 steps/sec

Example	Description
`SV2000000`	Run forward at 100 full steps/sec
`SV4000000`	Run forward at 200 full steps/sec
`SV-2000000`	Run reverse at 100 full steps/sec
`SV-1000000`	Run reverse at 50 full steps/sec
`SV0`	Decelerate to stop

Response:

```
> SV2000000
Setting velocity: 2000000 usteps/10000s  (~100 full steps/sec)
OK
```

Step Mode Reference (200-step/rev motor)

Mode	Microsteps/rev	SV value for 100 full steps/sec
0 — Full step	200	`SV1000000`
1 — 1/2 step (DEFAULT)	**400**	**`SV2000000`**
2 — 1/4 step	800	`SV4000000`
3 — 1/8 step	1600	`SV8000000`

Set Target Position (SP)

Sets the target position. Motor will move to this position and stop.

Format: `SP<position>`
Units: Microsteps. At default 1/2 step mode: **400 microsteps = 1 revolution**.

Example	Description
`SP0`	Move to position 0 (home)
`SP400`	Move 1 revolution from home
`SP800`	Move 2 revolutions from home
`SP-400`	Move 1 revolution in reverse from home
`SP10000`	Move to position 10000

Response:

```
> SP1000
Setting target position: 1000 usteps
OK
```

Read Status (SR)

Reads and displays the current stepper motor status.

Format: `SR`

Response:

```
> SR
--- Stepper Status ---
  Operation state: Normal
  Current position: 1000 usteps
  Target position:  1000 usteps
  Current velocity: 0 usteps/10000s  (~0 full steps/sec)
  Step mode:        1/2
---------------------
```

Operation States

State	Description
Reset	Controller was reset
De-energized	Motor coils are off
Soft error	Error condition exists
Waiting for ERR line	Waiting for error line signal
Starting up	Initializing
Normal	Operating normally

Set Current Position to Zero (SC)

Halts the motor and sets the current position register to 0. Useful for establishing a home reference.

Format: `SC`

Response:

```
> SC
Setting current position to: 0 usteps
OK
```

Set Max Speed (SM)

Sets the maximum allowed speed. Uses the same units as SV (microsteps per 10,000 seconds).

Format: `SM<speed>`

Example	Description
`SM2000000`	Max speed 100 full steps/sec (1/2 microstep mode)
`SM10000000`	Max speed 500 full steps/sec (1/2 microstep mode)

Set Max Acceleration (SA)

Sets the maximum acceleration and deceleration rate. Also sets deceleration to the same value.

Format: `SA<acceleration>`
Units: Microsteps per 100 s²

Example	Description
`SA10000`	Set moderate acceleration rate
`SA50000`	Set higher acceleration rate

Quick Reference Card

```
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VALVE COMMANDS
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1P<1-24>  Move valve 1 to position      2P<1-24>  Valve 2
1S        Query valve 1 status           2S        Valve 2
1M        Home valve 1                   2M        Valve 2
1E        Read valve 1 error             2E        Valve 2

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STEPPER COMMANDS
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SO        Stepper ON (energize + clear stale target)
SF        Stepper OFF (de-energize / coast)
SS        Stop (halt and hold position)
SR        Read stepper status
SC        Set current position to 0

SV<n>     Set velocity (microsteps/10000s)
          Positive = forward, Negative = reverse
          DEFAULT MODE: 1/2 microstepping — divide by 20,000 for full steps/sec
          Example: SV2000000 = 100 full steps/sec forward
          Example: SV-1000000 = 50 full steps/sec reverse
          Example: SV0 = decelerate to stop

SP<n>     Set target position (microsteps from 0)
          DEFAULT MODE: 400 microsteps = 1 revolution (200-step motor)

SM<n>     Set max speed (microsteps/10000s)
SA<n>     Set max acceleration (microsteps/100s²)

?         Show help
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```

Usage Examples

Example 1: Basic Valve Operation

```
1S        Query valve 1 current position
1M        Home valve 1
1P5       Move valve 1 to position 5
1S        Verify position
2P12      Move valve 2 to position 12
```

Example 2: Run Stepper at Constant Speed

```
SO        Turn stepper on (energize + clear stale target)
SR        Check status
SV2000000 Run forward at 100 full steps/sec
          (motor runs continuously)
SV0       Decelerate to stop
SF        Turn stepper off
```

Example 3: Run Stepper in Reverse

```
SO         Turn stepper on
SV-4000000 Run reverse at 200 full steps/sec
SV0        Decelerate to stop
```

Example 4: Move Stepper to Position

```
SO        Turn stepper on
SC        Set current position as 0 (home)
SP400     Move to 1 revolution from home
SR        Check if arrived
SP0       Return to home position
```

Example 5: Emergency Stop

```
SS        Immediately stop and hold position
```

Example 6: Complete System Sequence

```
SO        Enable stepper
1M        Home valve 1
2M        Home valve 2
1P3       Valve 1 to position 3
SV1000000 Run stepper at 50 full steps/sec
          (wait for desired time)
SV0       Stop stepper
1P5       Valve 1 to position 5
2P8       Valve 2 to position 8
SF        De-energize stepper
```

Troubleshooting

Valve Issues

Symptom	Possible Cause	Solution
No response	Wiring incorrect	Check TX→RXD, RX→TXD, GND
No response	No power	Verify 24 VDC to valve
Error 0x42	Position error	Home valve and retry
Error 0x63	Home failure	Check for obstruction

Stepper Issues

Symptom	Possible Cause	Solution
No response	Wiring incorrect	Check TX→RX, RX→TX, GND
Motor won't move	Not energized	Send `SO` command
Motor won't move	Safe start active	Send `SO` (includes exit safe start)
Motor won't move	Wrong control mode	Use Tic Control Center to set "Serial/I²C/USB" mode
Motor stutters	Speed too high	Reduce velocity or increase acceleration ramp
Wrong direction	Wiring reversed	Swap motor coil wires or use negative velocity
Unexpected motion on power-up	Stale target in controller	Firmware v2.1.7+ prevents this — keepalive only sent when motor is intentionally active. Ensure latest firmware is flashed.

Wiring Diagram

```
            ARDUINO MEGA
          ┌──────────────┐
          │              │
PC ──USB──┤ USB          │
          │              │
          │ Pin 18 TX1   ├──────► IDEX Valve 1 J2-10 (RXD)
          │ Pin 19 RX1   ◄──────┤ IDEX Valve 1 J2-9  (TXD)
          │ GND          ├──────► IDEX Valve 1 J2-4  (GND)
          │              │
          │ Pin 16 TX2   ├──────► IDEX Valve 2 J2-10 (RXD)
          │ Pin 17 RX2   ◄──────┤ IDEX Valve 2 J2-9  (TXD)
          │ GND          ├──────► IDEX Valve 2 J2-4  (GND)
          │              │
          │ Pin 14 TX3   ├──────► Tic T500 RX
          │ Pin 15 RX3   ◄──────┤ Tic T500 TX
          │ GND          ├──────► Tic T500 GND
          │              │
          └──────────────┘
```